

Lab 2: Feedback Control of a Ping-Pong Ball

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Group 3

1 Exercise 1

1.1 Circuit

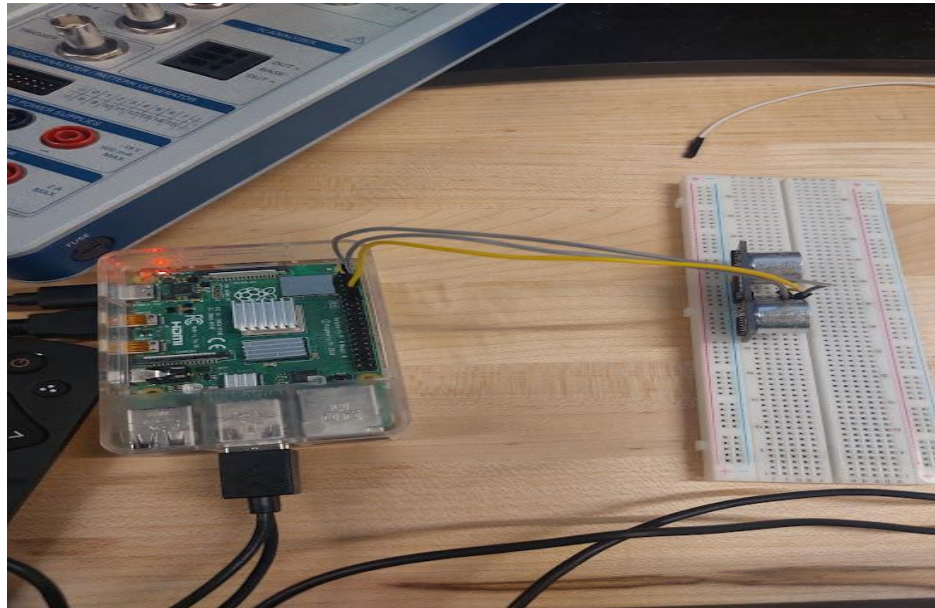


Figure 1: Circuit for exercise 1

1.2 Summary

In this exercise, we utilized a sonar sensor and Raspberry Pi to create a circuit. The code read the distance measurements from the sensor with a 5 μ s pulse, converting the distance to centimeters, and printing the information to the screen.

2 Exercise 2

2.1 Circuit

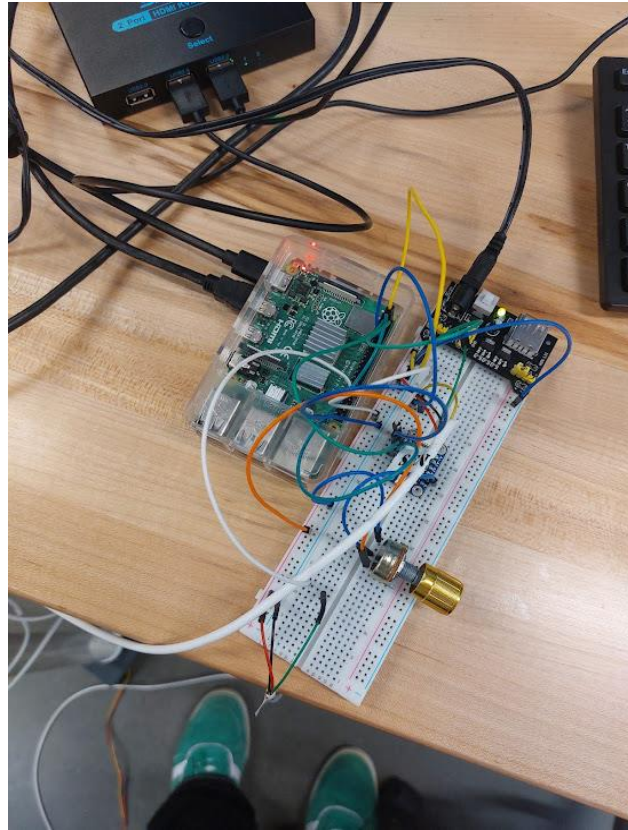


Figure 2: Circuit for exercise 2



Figure 3 Piing Pong setup for exercise 2

2.2 Summary

The circuit was constructed using ADS1015, sonar sensor, potentiometer, fan and power supply. The sensor and fan were connected to a ping-pong ball inside a tube. Software was created to control the fan with the PID and change speeds to adjust the height from the sonar sensor.

Supplemental Questions

1. Briefly summarize what you learned from this lab.

In this lab we learned how sonar works and how to use it to measure distance. We first tested how the sonar sensor worked, and later integrated it with a fan. We also learned PID algorithm control and how to use it to maintain desired output, staying at a certain height in this case.

2. In PID control, how will the values of the P, I and D parameters affect your control performance?

The proportional part controls the steady state error, as it increases the steady-state error decreases. The Integral part further helps remove steady-state error of the proportional by accumulating the errors, and it drives the variables to the desired setpoint. As the P increases there are unwanted oscillations and that is where the Derivative helps smooth them out, as D increases oscillation decreases.

3. How did you decide the value of the P, I, D parameters to achieve a good control performance?

We initially used the values $K_p = 100$, $K_i = 200$, $K_d = 20$, and using trial and error we adjusted the values. We utilized the following table in our assessment of what would be the best K_p , K_i , and K_d values.

	rise time	overshoot	settling time	steady state error
K_p	decreases	increases	minor effect	decreases
K_i	decreases	increases	increases	eliminates
K_d	minor effect	decreases	decreases	minor effect

Figure 4 Lecture 5 PID Control Parameters

We settled on the values: $K_p = 2$, $K_i = 0.0005$, $K_d = 2$. Because these parameters needed to be responsive to the object distance, we used if statements to attenuate the K_p , K_i and K_d values.

ACKNOWLEDGMENTS

I certify that this report is my/our own work, based on my/our personal study and/or research and that I/we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. I/We also certify that this assignment/report has not previously been submitted for assessment anywhere, except where specific permission has been granted from the coordinators involved.

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